

Markov Chains

Markov Chain

A Markov chain is specified by a transition matrix $P \in \mathbb{R}^{n \times n}$:

- ▶ $P(X_{t+1} = i \mid P(X_t = j) = P_{ij} \geq 0$
- ▶ $\sum_i P_{ij} = 1$ (column stochastic)

The stationary distribution of the Markov chain (if it exists) is an eigenvector π of the transition matrix P with eigenvalue 1:

$$P\pi = \pi, \quad \pi_i \geq 0, \quad \sum_i \pi_i = 1$$

Irreducibility and Aperiodicity

A Markov chain is called **irreducible** if for any $i, j \in \Sigma$, there exists $t \in \mathbb{N}$ such that:

$$(P^t)_{ij} > 0$$

Define the set of return times to state i :

$$T(i) := \{t \geq 1 : (P^t)_{ii} > 0\}$$

The **period** of i is:

$$\gcd T(i)$$

A Markov chain is **aperiodic** if every state has period 1.

Perron's Theorem

Let $P \in \mathbb{R}^{n \times n}$ be the transition matrix of a finite, irreducible Markov chain. Then:

- ▶ 1 is an eigenvalue of P
- ▶ The eigenvalue 1 is simple (i.e., has multiplicity 1)
- ▶ There exists a unique (up to scaling) eigenvector π with strictly positive entries such that:

$$P\pi = \pi$$

- ▶ All other eigenvalues λ satisfy:

$$|\lambda| < 1$$

Proposition

Let P be the transition matrix of a finite, irreducible and aperiodic Markov chain. Then it has a unique stationary distribution π with eigenvalue 1. Moreover, there exists $\gamma \in (0, 1]$ such that every eigenvalue $\lambda \neq 1$ of P satisfies

$$|\lambda| \leq 1 - \gamma.$$

Proof of Proposition

From the assumptions, there exists $t \in \mathbb{N}$ such that P^t is a positive matrix. Let π be the eigenvector of P^t corresponding to the unique maximal eigenvalue with positive entries. Since P^t is column-stochastic,

$$\sum_{i,j} (P^t)_{ij} \pi_j = \sum_j \pi_j. \quad (1)$$

Therefore the corresponding eigenvalue is 1. We may normalize π so that $\sum_i \pi_i = 1$.

From

$$P^t(P\pi) = P(P^t\pi), \quad (2)$$

we find that $P\pi$ is also an eigenvector of P^t with eigenvalue 1. By the uniqueness of the eigenvector, we have $P\pi = \pi$, i.e., π is the unique stationary distribution of P .

Proof of Proposition (contd.)

By the Perron theorem, all other eigenvalues of P^t have absolute value strictly less than 1, so there exists $\gamma' > 0$ such that they are all bounded in absolute value by $1 - \gamma'$.

Therefore if there is an eigenvector v of P with eigenvalue $\lambda \neq 1$, then v is an eigenvector of P^t and

$$|\lambda|^t \leq 1 - \gamma'.$$

The result then follows once we define

$$1 - \gamma = (1 - \gamma')^{1/t}.$$

Convergence Theorem

For irreducible, aperiodic chains:

$$\lim_{t \rightarrow \infty} P^t \pi^{(0)} = \pi$$

Mixing time depends on spectral gap γ :

$$\gamma = 1 - \lambda_2$$

Complexity:

$$O(1/\gamma)$$

Proof sketch

Write $\pi^{(0)}$ as a linear combination of the orthonormal eigenvectors q_i of P ($q_1 = \pi$):

$$\pi^{(0)} = a_1 q_1 + a_2 q_2 + \cdots + a_n q_n.$$

Since $\pi^{(t)}$ is a multiple of $P^t \pi^{(0)}$, we have for some constants c_t ,

$$\begin{aligned}\pi^{(t)} &= c_t P^t \pi^{(0)} \\ &= c_t (a_1 \lambda_1^t q_1 + a_2 \lambda_2^t q_2 + \cdots + a_n \lambda_n^t q_n) \\ &= c_t \lambda_1^t \left(a_1 q_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^t q_2 + \cdots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^t q_n \right).\end{aligned}$$

Suppose $q_1^T v^{(0)} \neq 0$. Then

$$\|v^{(k)} - (\pm q_1)\| = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right) = O((1-\gamma)^k) \approx O(e^{-\gamma k}) \Rightarrow k \geq \frac{1}{\gamma} \log \frac{1}{\varepsilon}$$

Definition: Reversible Markov Chain

A Markov chain is reversible if there is a stationary distribution π that satisfies the **detailed balance** condition:

$$\pi_i P_{ij} = \pi_j P_{ji} \quad (3)$$

Example

Metropolis-type construction:

- ▶ Target distribution: $\pi_i \propto e^{-\beta E(i)}$
- ▶ Transition:

$$P_{ij} = Q_{ij} \min\{1, e^{-\beta(E(j)-E(i))}\}$$

Satisfies detailed balance.

Discriminant Matrix

$$D := \sum_{i,j} \sqrt{P_{ij}P_{ji}} |i\rangle\langle j| \quad (4)$$

Properties:

- ▶ Symmetric $\Rightarrow$ Hermitian
- ▶ Encodes spectral information of P

Proposition

For finite, aperiodic, irreducible, reversible chains:

$$D|\pi\rangle = |\pi\rangle \quad (5)$$

Similarity relation:

$$D = \text{diag}(\sqrt{\pi})^{-1} P \text{diag}(\sqrt{\pi})$$

Consequences:

- ▶ P and D share eigenvalues
- ▶ P is diagonalizable

Key Takeaways

- ▶ Irreducibility $\Rightarrow$ unique stationary distribution
- ▶ Reversibility $\Rightarrow$ symmetric structure via D
- ▶ Spectral gap controls convergence